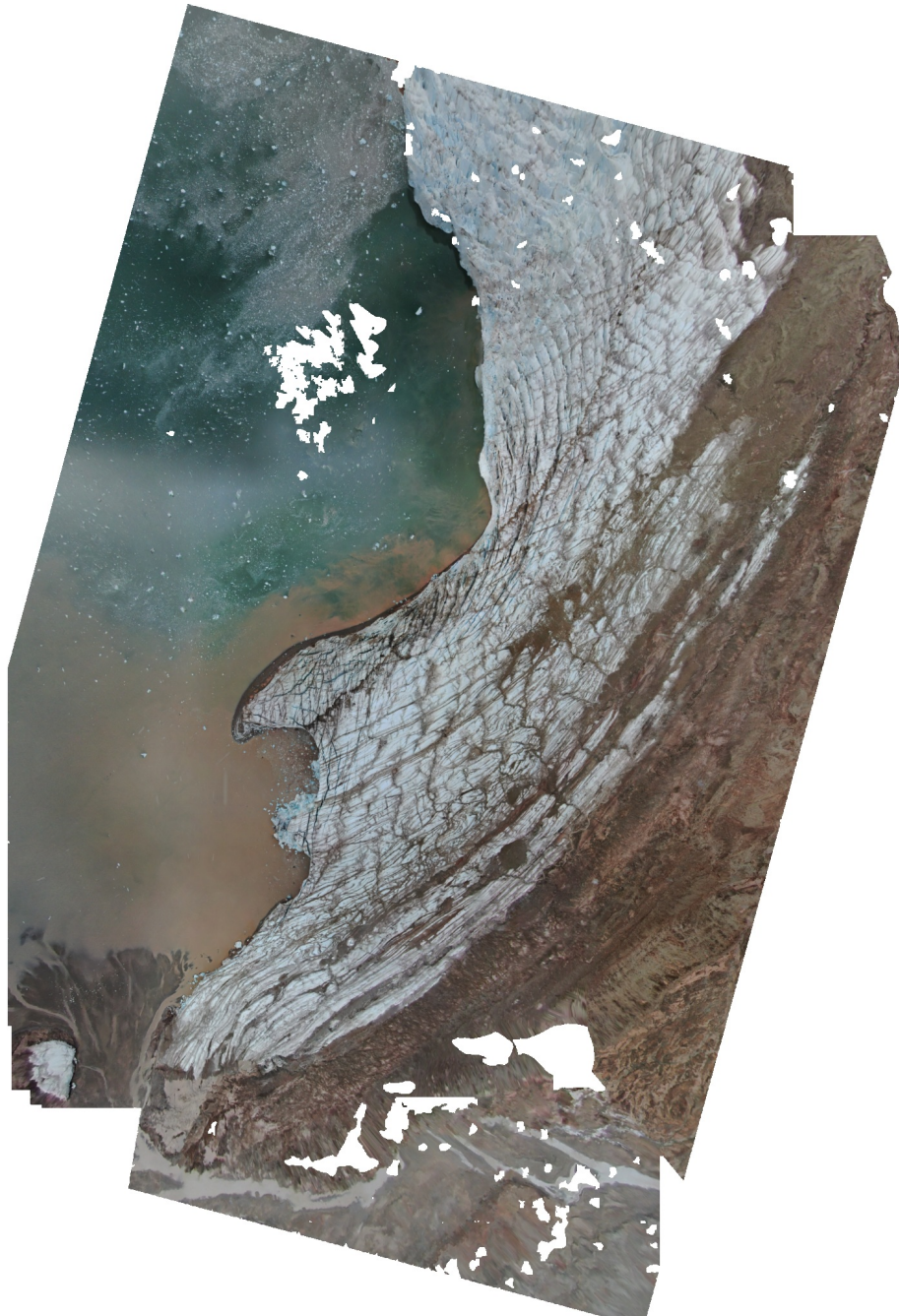


# Agisoft Metashape

Processing Report  
15 September 2026



# Survey Data

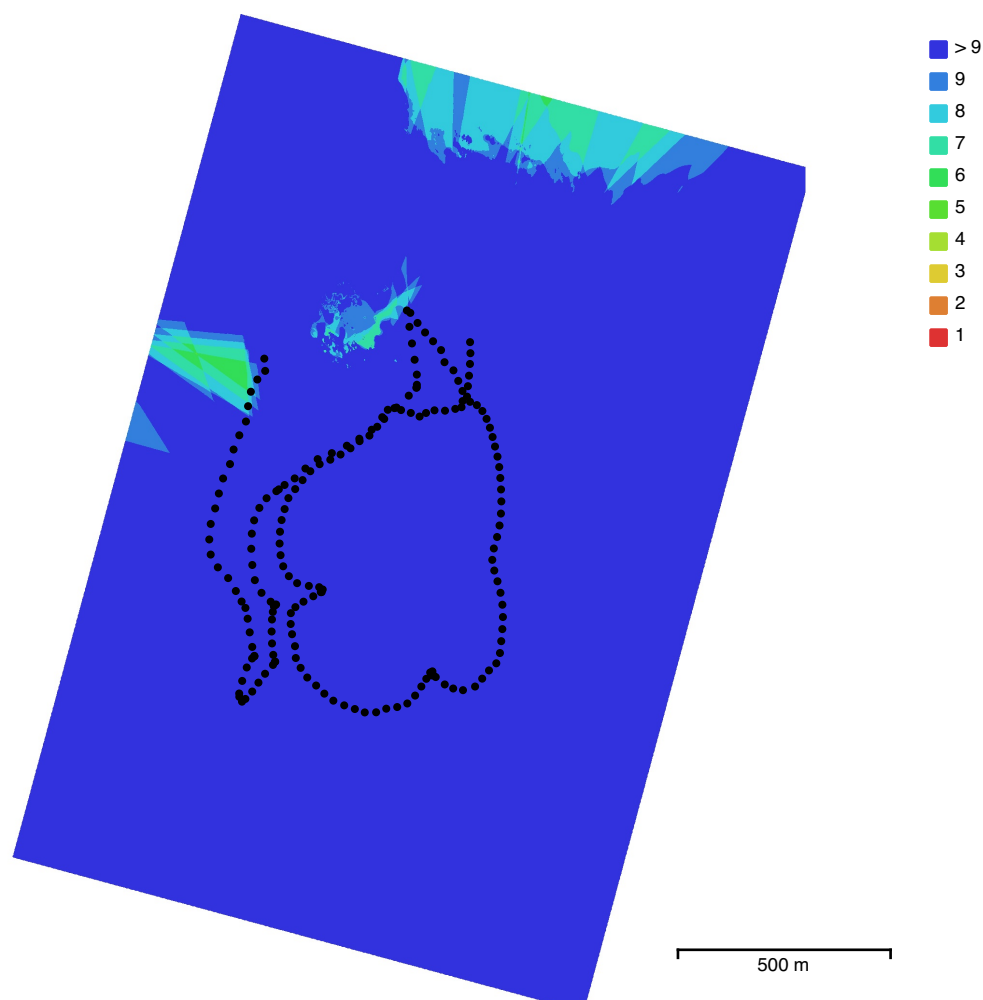


Fig. 1. Camera locations and image overlap.

|                    |             |                     |         |
|--------------------|-------------|---------------------|---------|
| Number of images:  | 188         | Camera stations:    | 188     |
| Flying altitude:   | 259 m       | Tie points:         | 79,982  |
| Ground resolution: | 10.5 cm/pix | Projections:        | 439,474 |
| Coverage area:     | 2.85 km²    | Reprojection error: | 1.1 pix |

| Camera Model       | Resolution  | Focal Length | Pixel Size     | Precalibrated |
|--------------------|-------------|--------------|----------------|---------------|
| Test_Pro (4.386mm) | 4000 x 3000 | 4.386 mm     | 1.58 x 1.58 µm | No            |

Table 1. Cameras.

# Camera Calibration

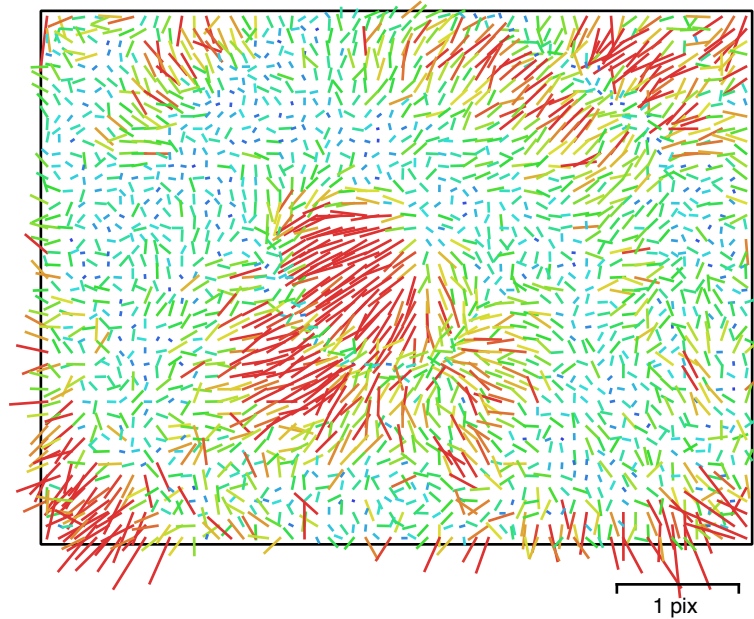


Fig. 2. Image residuals for Test\_Pro (4.386mm).

## Test\_Pro (4.386mm)

188 images

|       |             |              |                           |
|-------|-------------|--------------|---------------------------|
| Type  | Resolution  | Focal Length | Pixel Size                |
| Frame | 4000 x 3000 | 4.386 mm     | 1.58 x 1.58 $\mu\text{m}$ |

|    | Value       | Error   | F    | Cx    | Cy    | K1    | K2    | K3    | P1    | P2    |
|----|-------------|---------|------|-------|-------|-------|-------|-------|-------|-------|
| F  | 2872.46     | 0.084   | 1.00 | -0.07 | -0.68 | -0.11 | 0.18  | -0.17 | -0.07 | -0.14 |
| Cx | -10.185     | 0.11    |      | 1.00  | 0.06  | 0.02  | -0.02 | 0.02  | 0.95  | 0.10  |
| Cy | -24.6907    | 0.11    |      |       | 1.00  | -0.07 | 0.04  | -0.03 | 0.05  | 0.51  |
| K1 | -0.0325491  | 7e-05   |      |       |       | 1.00  | -0.96 | 0.90  | 0.02  | -0.09 |
| K2 | 0.0430997   | 0.00022 |      |       |       |       | 1.00  | -0.98 | -0.02 | 0.05  |
| K3 | -0.0339766  | 0.00021 |      |       |       |       |       | 1.00  | 0.01  | -0.05 |
| P1 | -0.00133778 | 1.1e-05 |      |       |       |       |       |       | 1.00  | 0.08  |
| P2 | -0.00205463 | 6.8e-06 |      |       |       |       |       |       |       | 1.00  |

Table 2. Calibration coefficients and correlation matrix.

# Camera Locations

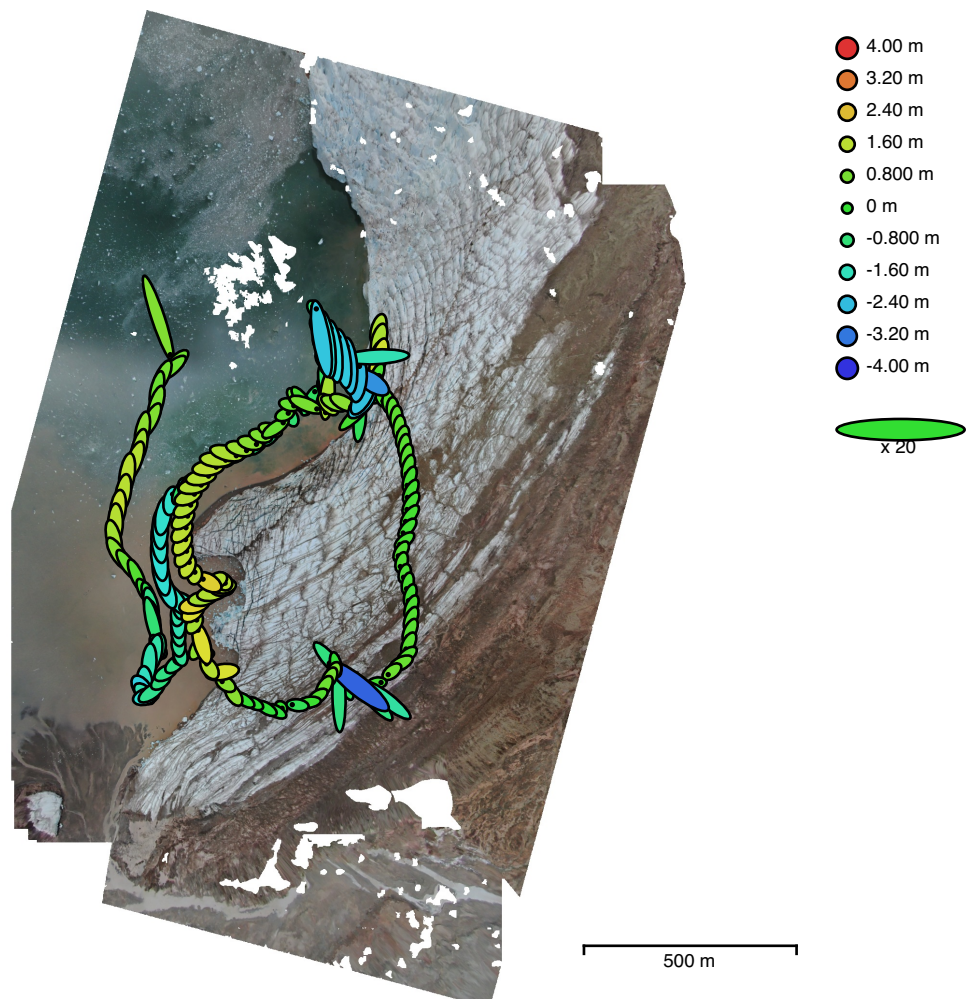


Fig. 3. Camera locations and error estimates.  
Z error is represented by ellipse color. X,Y errors are represented by ellipse shape.  
Estimated camera locations are marked with a black dot.

| X error (m) | Y error (m) | Z error (m) | XY error (m) | Total error (m) |
|-------------|-------------|-------------|--------------|-----------------|
| 1.23344     | 2.07849     | 1.2164      | 2.41692      | 2.70576         |

Table 3. Average camera location error.  
X - Longitude, Y - Latitude, Z - Altitude.

# Digital Elevation Model

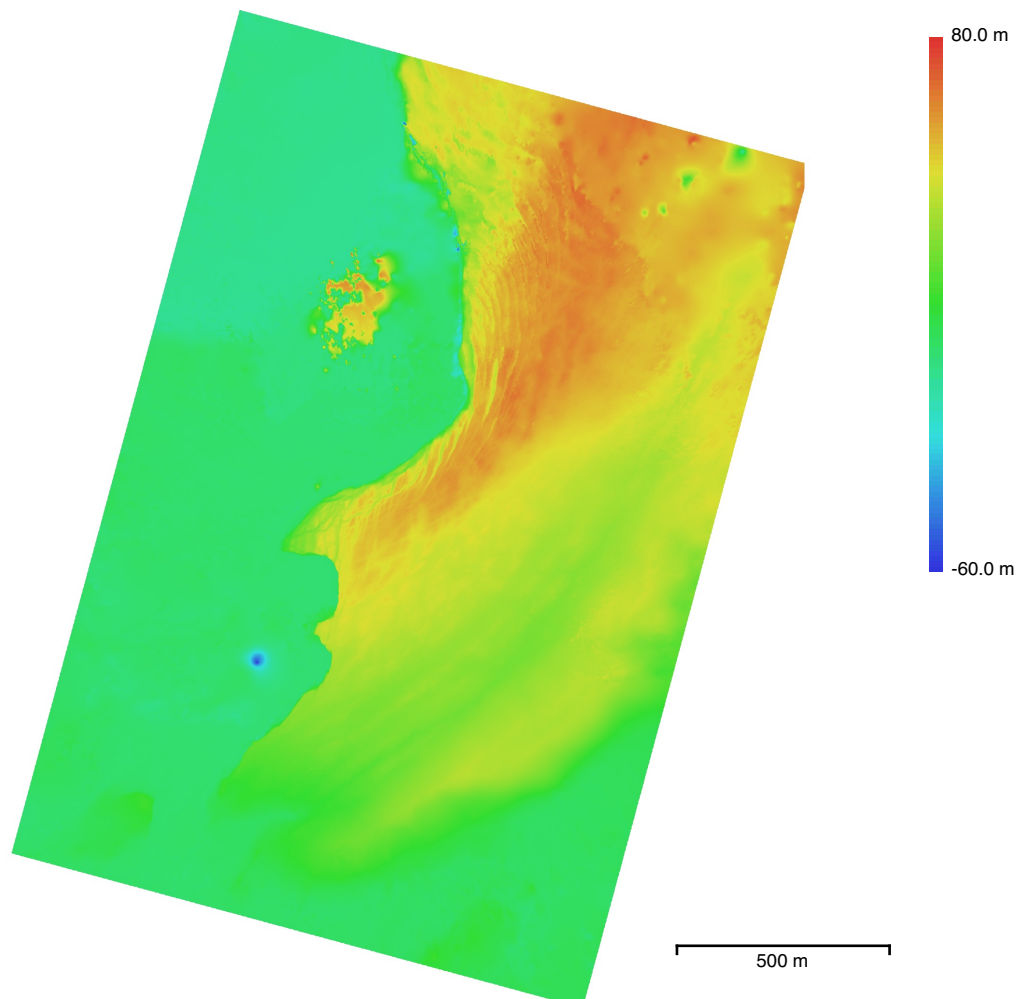


Fig. 4. Reconstructed digital elevation model.

Resolution: 21.1 cm/pix  
Point density: 22.6 points/m<sup>2</sup>

# Processing Parameters

## General

|                   |                     |
|-------------------|---------------------|
| Images            | 188                 |
| Aligned images    | 188                 |
| Coordinate system | WGS 84 (EPSG::4326) |
| Rotation angles   | Yaw, Pitch, Roll    |

## Tie Points

|                                |                        |
|--------------------------------|------------------------|
| Points                         | 79,982 of 128,721      |
| RMS reprojection error         | 0.262211 (1.09947 pix) |
| Max reprojection error         | 0.818824 (35.548 pix)  |
| Mean key point size            | 3.74769 pix            |
| Point colors                   | 3 bands, uint8         |
| Key points                     | No                     |
| Average tie point multiplicity | 6.51798                |

## Alignment parameters

|                               |                      |
|-------------------------------|----------------------|
| Accuracy                      | High                 |
| Generic preselection          | Yes                  |
| Reference preselection        | Source               |
| Key point limit               | 40,000               |
| Key point limit per Mpx       | 1,000                |
| Tie point limit               | 4,000                |
| Exclude stationary tie points | Yes                  |
| Guided image matching         | No                   |
| Adaptive camera model fitting | No                   |
| Matching time                 | 3 minutes 56 seconds |
| Matching memory usage         | 8.22 GB              |
| Alignment time                | 46 seconds           |
| Alignment memory usage        | 640.75 MB            |
| Date created                  | 2026:09:11 11:37:53  |
| Software version              | 2.3.2.22956          |
| File size                     | 15.03 MB             |

## Depth Maps

|                                         |                      |
|-----------------------------------------|----------------------|
| Count                                   | 188                  |
| <b>Depth maps generation parameters</b> |                      |
| Quality                                 | High                 |
| Filtering mode                          | Mild                 |
| Max neighbors                           | 16                   |
| Processing time                         | 3 minutes 49 seconds |
| Date created                            | 2026:09:11 11:45:00  |
| Software version                        | 2.3.2.22956          |
| File size                               | 678.61 MB            |

## Point Cloud

|                      |            |
|----------------------|------------|
| Points               | 40,131,684 |
| Coordinate precision | 5.26 cm    |

## Point attributes

|        |                |
|--------|----------------|
| Color  | 3 bands, uint8 |
| Normal |                |

## Point classes

|                            |            |
|----------------------------|------------|
| Created (never classified) | 40,131,684 |
|----------------------------|------------|

## Depth maps generation parameters

|                 |                      |
|-----------------|----------------------|
| Quality         | High                 |
| Filtering mode  | Mild                 |
| Max neighbors   | 16                   |
| Processing time | 3 minutes 49 seconds |

## Point cloud generation parameters

|             |            |
|-------------|------------|
| Source data | Depth maps |
|-------------|------------|

|                                  |                                |
|----------------------------------|--------------------------------|
| Processing time                  | 48 minutes 50 seconds          |
| Date created                     | 2026:09:11 12:34:35            |
| Software version                 | 2.3.2.22956                    |
| File size                        | 542.78 MB                      |
| <b>DEM</b>                       |                                |
| Size                             | 8,845 x 11,035                 |
| Resolution                       | 21.1 cm/pix                    |
| Coordinate system                | WGS 84 (EPSG::4326)            |
| <b>Reconstruction parameters</b> |                                |
| Source data                      | Point cloud                    |
| Interpolation                    | Enabled                        |
| Processing time                  | 14 seconds                     |
| Memory usage                     | 334.41 MB                      |
| Date created                     | 2026:09:11 12:50:20            |
| Software version                 | 2.3.2.22956                    |
| File size                        | 330.12 MB                      |
| <b>Orthomosaic</b>               |                                |
| Size                             | 17,693 x 22,070                |
| Resolution                       | 10.5 cm/pix                    |
| Coordinate system                | WGS 84 (EPSG::4326)            |
| Colors                           | 3 bands, uint8                 |
| Orthophotos                      | 1.30 GB                        |
| <b>Reconstruction parameters</b> |                                |
| Blending mode                    | Mosaic                         |
| Surface                          | DEM                            |
| Enable hole filling              | Yes                            |
| Enable ghosting filter           | No                             |
| Processing time                  | 25 minutes 18 seconds          |
| Memory usage                     | 3.87 GB                        |
| Date created                     | 2026:09:11 13:13:58            |
| Software version                 | 2.3.2.22956                    |
| File size                        | 1.48 GB                        |
| <b>System</b>                    |                                |
| Software name                    | Agisoft Metashape Professional |
| Software version                 | 2.3.2 build 22956              |
| OS                               | Mac OS 64 bit                  |
| RAM                              | 16.00 GB                       |
| CPU                              |                                |
| GPU(s)                           | Apple M4                       |